An Attracting 3-adic Schröder Germ with Unbounded Denominator Powers

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Abstract

Conjecture 00000000746 asserts that the convergent Schröder function of a p-adic attracting fixed point has coefficient denominators involving at most the first power of p. We give an explicit normalized analytic local coordinate over $\mathbb{Q}_3$ with coefficient $1/9$ at degree three, and more generally coefficient 3^{-n} at degree $n+1$. We verify the invariant neighborhood, analytic germs, nonzero attracting multiplier, all orbit limits, the Schröder identity, actual convergent coefficient series and analytic inverse. A pinned Lean project proves these properties for actual p-adic numbers and the necessary denominator assertion's negation. Only the denominator clause is refuted, not analytic linearization.

1 Exact statement and conventions

Both original languages say that the Schröder function of an attracting fixed point is a convergent power series whose coefficient denominators contain only the first power of p. We use the standard local analytic-germ meaning of the named p-adic Koenigs/Schröder function: at a fixed point moved to zero, an analytic coordinate H satisfies $H(0)=0$, $H'(0)=1$ and $H(F(x))=\lambda H(x)$, where $0<|\lambda|_p<1$. Our coordinate has an explicit analytic inverse, so it is not merely a formal eigenfunction.

The supplied statement imposes no integral-coefficient input condition, polynomial input, global absence of poles, or preassigned unit disk. Adding one would change the conjecture. All analytic claims below are local on the explicitly stated open neighborhood. For our rational coefficients in lowest terms, the denominator clause necessarily implies

$$p^2 \nmid \operatorname{den}(a_m) \quad \text{for every } m \geq 0. \tag{1}$$

This is the exponent-at-most-one reading; the stronger reading that the only denominators are 1 and p also implies it. The value $1/9$ violates both at $p=3$. The exact unchanged bilingual source and its Git/SHA-256 identities accompany the submission.

2 Actual analytic maps and their neighborhood

Work in the complete non-Archimedean normed field $K=\mathbb{Q}_3$, with $|3|_3=1/3$, and set

$$\mathcal{U}=\{x\in K : |x|_3<1/3\}, \qquad F(x)=\frac{3x}{1+2x/3}, \quad H(x)=\frac{x}{1-x/3}, \quad J(x)=\frac{x}{1+x/3}.$$

The set $\mathcal{U}$ is open, nonempty and contains zero. For every $x\in\mathcal{U}$, $|x/3|_3<1$, and $|2|_3\leq 1$ gives $|2x/3|_3<1$. The strong ultrametric equality yields

$$|1-x/3|_3=|1+2x/3|_3=|1+x/3|_3=1.$$

Thus all denominators are nonzero throughout $\mathcal{U}$, and the three rational maps are K-analytic on neighborhoods of every point in $\mathcal{U}$. Total division outside that set in Lean's field operations does not assert global analyticity at poles.

We have $F(0)=H(0)=0$, $F'(0)=3\neq 0$, $H'(0)=1$, and $|F'(0)|_3=1/3<1$. These are actual derivatives over K, not derivatives over an unrelated real scalar field. Moreover, for all $x\in\mathcal{U}$,

$$|F(x)|_3=|x|_3/3, \qquad |H(x)|_3=|J(x)|_3=|x|_3. \tag{2}$$

In particular all three maps preserve $\mathcal{U}$. Induction on the actual iterates of F gives

$$|F^{\circ n}(x)|_3 = (1/3)^n |x|_3 \longrightarrow 0 \quad \text{for every } x \in \mathcal{U}.$$

The fixed point is genuinely attracting, not just a point assigned a contracting multiplier.

3 Normalized analytic linearization and inverse

Writing $A = 1 + 2x/3$, direct algebra gives $1 - F(x)/3 = (1 - x/3)/A$. The already proved unit denominators therefore give

$$H(F(x)) = 3H(x) \qquad (x \in \mathcal{U}). \tag{3}$$

Similarly $1 - J(x)/3 = 1/(1 + x/3)$ and $1 + H(x)/3 = 1/(1 - x/3)$, so

$$H(J(x)) = x, \qquad J(H(x)) = x \qquad (x \in \mathcal{U}).$$

Together with analyticity, normalization and preservation of $\mathcal{U}$, these equations exhibit an actual analytic conjugacy between the germ of F and multiplication by 3. No unrestricted Koenigs existence theorem is assumed as a new Lean axiom.

The local analytic interpretation is also standard in the literature; see Jenkins–Spallone, *A p-adic approach to local analytic dynamics: analytic conjugacy of analytic maps tangent to the identity*, Annales de la Faculté des Sciences de Toulouse 18 (2009), 611–634, doi:10.5802/afst.1217. Our explicit identities prove this instance directly; the reference does not supply unproved premises or change the original conjecture.

4 The actual convergent series and rational denominators

For every $x \in \mathcal{U}$, the geometric-series criterion $|x/3|_3 < 1$ proves

$$H(x) = x \sum_{n=0}^{\infty} (x/3)^n = \sum_{n=0}^{\infty} 3^{-n} x^{n+1}.$$

Indexing by the actual exponent of x, let

$$a_0 = 0, \qquad a_{n+1} = 3^{-n} \in \mathbb{Q}.$$

The coefficient sequence embedded in K genuinely sums to $H(x)$ at *every* point of $\mathcal{U}$. Lean proves both `HasSum` and `HasFPowerSeriesAt` for the formal multilinear series `ofScalars` of these coefficients, and identifies its actual coefficient at each degree. Thus a_3 is not an unrelated numerical label attached to H.

The canonical reduced denominator of 3^{-n} is 3^n, proved using the actual rational normal form. In particular

$$a_1 = 1, \quad a_2 = 1/3, \quad a_3 = 1/9, \qquad \operatorname{den}(a_3) = 9 = 3^2.$$

Genuine analytic-series uniqueness additionally proves that any rational coefficient sequence giving the same analytic germ H has these same coefficients. The code does not advertise a general theorem of uniqueness of all normalized Schröder germs.

Theorem 1. *The normalized analytic Schröder coordinate H of the attracting fixed point of F violates the first-power denominator assertion of conjecture 00000000746.*

Proof. The preceding arguments establish every actual analytic, fixed-point, normalization, invariance, attraction and functional-equation premise. At degree three the actual coefficient has denominator 9, divisible by 3^2, contrary to (1). In fact the denominators 3^n are unbounded. The convergent-series part of the statement continues to hold in this counterexample; it is only the denominator clause that fails. □

5 Formal correspondence and reproducibility

The transparent `NormalizedAnalyticSchroder` predicate includes a nonempty open neighborhood containing zero, K-analytic maps, actual fixed point and derivatives, nonzero attracting multiplier, invariant set, exact norm contraction, convergence of every actual orbit, the Schröder identity and both actual series specifications. `actual_normalized_Schroder` proves its concrete instance; `exists_actual_`

`counterexample` supplies an actual witness with the failed bound; `necessary_denominator_claim_false` negates the corresponding necessary universal assertion without leaving a premise to discharge. The explicit inverse and analytic conjugacy are verified separately, strengthening the standard coordinate interpretation.

Analysis uses the ordinary inherited field, self-module and normed-space operations on $\mathbb{Q}_3$. `self_smul_eq_mul` checks that scalar action is the usual multiplication; no topology, norm or mathematical assumption is changed to make a theorem compile. The formal scope is this genuine explicit counterexample, not a hard-coded Boolean or a corrected conjecture about integral-input or global maps.

The project pins Lean 4.33.1 and Mathlib commit `0df444a360eaa60ab8c11dca51a86af692955474`, with a portable manifest fixing all transitive Git revisions. The Python reproducer verifies source bytes, the exact toolchain, every dependency HEAD/origin and clean tracked source, deletes only this project's generated build cache, rebuilds with warnings as errors and audits all 45 public theorems plus the prime instance. Only `propext`, `Classical.choice` and `Quot.sound` are allowed. Exact rational sample checks are supplemental regressions, not substitutes for the universal p-adic analytic proofs. The actual PDF is compiled from this source by Tectonic 0.15.0 with `SOURCE_DATE_EPOCH=0`.

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